

Dashboards overview

Three dashboards linked via top-bar navigation buttons. All share the same `asset` / `symbol` template variables and time range.

Dashboard	UID	Nav label	Purpose
Inventory (main)	<code>metrics-margin-main</code>	—	Margin pool health: stress proxy, inventory, correlation, risk config tables
Stacked View	<code>metrics-margin-stacked</code>	Stacked View	Same panels as Inventory but full-width stacked layout
Flow	<code>metrics-margin-pibrbs</code>	Flow	Money flow analytics (taker volume, net flow) and PIBRBS overlay charts

From Inventory/Stacked you can reach Flow via the **Flow** button. From Flow you can reach Inventory via the **Inventory** button.

Inventory dashboard layout

Rows:

- `Overview`
- `Correlation`
- `Margin Pool Proxy`
- `Risk / Config Tables`

Includes:

- spot price over time
- margin available inventory over time
- inverted borrow-pressure proxy
- rolling 24h and 7d correlation
- normalized comparison panel
- price index table
- isolated tier table
- collateral ratio table
- risk-based liquidation table
- config change events table
- annotations sourced from `config_change_events`

Inventory dashboard components

Overview row (stat panels)

Panel	What it shows	Why it matters
Latest Spot Price	Current close price from 5-min klines	Baseline reference for all other indicators
Latest Available Inventory	Current margin pool supply for the asset	Absolute amount available for borrowing on the exchange
Current Stress Proxy	Negative z-score of inventory (higher = more stress)	Quantifies how far below normal the pool supply is. >1.5 = high stress, >0.5 = medium

Panel	What it shows	Why it matters
24h Inventory Change %	Pool supply change over the last 24 hours	Fast-moving signal for sudden borrow demand spikes
7d Inventory Change %	Pool supply change over the last 7 days	Trend context — persistent drawdown is more significant than a single-day drop
Stress Regime	Categorical label: LOW / MEDIUM / HIGH	Quick glance risk assessment derived from the stress proxy value

Correlation row (time series)

Panel	What it shows
Asset Price	Spot kline close prices over the selected time range
Margin Available Inventory	Margin pool supply over time
Inverted Borrow-Pressure Proxy	Stress proxy as bar chart — positive bars = below-normal inventory (borrow pressure), negative bars = above-normal (slack)
Rolling Correlation	24h and 7d rolling Pearson correlation between price returns and inventory changes. Range: -1 to +1

Margin Pool Proxy row

Panel	What it shows
Normalized Comparison (Base 100)	Price and inventory both rebased to 100 at the start of the visible window. Makes divergence/convergence visually obvious on a shared scale

Isolated vs Cross Coverage row

Two time-series panels showing Binance-configured capacity ceilings alongside live cross-margin pool data.

Panel	What it shows
Isolated Margin Max Borrowable Capacity (Highest Tier)	Two constant reference lines: <code>isolated_base_capacity</code> (dashed blue, left axis) and <code>isolated_quote_capacity</code> (solid orange, right axis), both from the highest configured isolated-margin tier
Cross-Margin Inventory vs Isolated Max Borrowable	Live <code>cross_inventory</code> (orange, left axis) alongside the <code>isolated_base_capacity</code> ceiling (green, left axis) — both in base-asset units so they share one scale

What Base Max Borrowable and Quote Max Borrowable mean

These are **Binance-wide, pool-level lending ceilings** — they are **not per-account limits**. They represent the maximum a single borrower can draw at the highest isolated-margin tier for this specific trading pair. Binance sets and occasionally revises them as a risk management control over the lending pool.

Field	Asset	What is being bounded	Who can borrow it	Why
Base Max Borrowable	Base asset (e.g. BTC)	How much BTC the pool can lend per position	Traders wanting to short (borrow BTC → sell it)	BTC borrowed and sold creates short exposure

Field	Asset	What is being bounded	Who can borrow it	Why
Quote Max Borrowable	Quote asset (e.g. USDC)	How much USDC the pool can lend per position	Traders wanting to long (borrow USDC → buy BTC)	USDC borrowed and deployed buys the base

Lines are **straight (constant)** by design — they only change when Binance revises its risk parameters, which is infrequent.

How to use these for correlation and signal analysis

1. Pool Utilization Ratio (**cross_inventory / isolated_base_capacity**) Shown visually in the "Cross-Margin vs Isolated Max Borrowable" panel. When **cross_inventory** approaches **isolated_base_capacity**, the cross-margin pool is at or near the configured ceiling — meaning nearly all available supply is already lent out. A ratio close to 1 with a falling cross_inventory = pool draining toward the max. This is the most stressed state.
2. Capacity reduction as a leading warning When Binance lowers **isolated_base_capacity** or **isolated_quote_capacity**, it is **tightening lending willingness** — typically because they see increased risk in that asset. A capacity cut visible as a step-down on the reference line (and logged in Config Change Events) historically precedes elevated volatility.
3. Asymmetric capacity as directional bias If Binance raises **isolated_quote_capacity** (USDC lending ceiling) without raising **isolated_base_capacity** (BTC lending ceiling), they are implicitly expanding long-side capacity while capping short-side capacity — a subtle bullish bias in the risk model. The reverse signals short-side concern.
4. Cross/Isolated gap as leverage buffer The gap between **cross_inventory** (live, changing) and **isolated_base_capacity** (ceiling, fixed) is the remaining headroom in the lending pool relative to the configured maximum. A narrowing gap at the same time as a rising stress proxy = confirming signal of pool stress.

Risk / Config Tables row

Panel	What it shows
Margin Price Index	Recent priceIndex snapshots — the reference price Binance uses for margin calculations
Isolated Margin Tier	Leverage tiers with risk ratios and max borrowable amounts per tier
Cross Margin Collateral Ratio	Collateral discount rates per asset group — determines effective collateral value
Risk-Based Liquidation Ratio	Liquidation and warning thresholds — changes here signal exchange risk parameter shifts
Configuration Change Events	Timestamped log of any changes detected in margin config endpoints

Annotations

Red vertical markers on all time series panels indicate **config change events** — moments when Binance changed margin tiers, collateral ratios, or liquidation parameters.

Flow dashboard (PIBRBS)

Accessible via the **Flow** link from the Inventory or Stacked dashboards. Contains four collapsible sections derived from **spot_klines** and **margin_available_inventory_snapshots** data. A hidden **quote_asset** template variable auto-resolves the quote currency from the selected symbol (e.g. USDC for BTCUSDC).

Money Flow Lite row

Built from `spot_klines` fields: `taker_buy_base_volume`, `volume`, and `close_time`. All volume metrics use base-asset units. "Sell volume" is derived as `volume - taker_buy_base_volume`.

Stat panels (top three rows)

Panel	What it shows	Calculation
Buy Volume (5m candle) Latest	Taker buy volume of the most recent candle	<code>taker_buy_base_volume</code> of last row
Sell Volume (5m candle) Latest	Taker sell volume of the most recent candle	<code>volume - taker_buy_base_volume</code> of last row
Buy Share % (5m candle) Latest	Buyer share of total volume in the most recent candle	<code>taker_buy_base_volume / volume</code>
Net Taker Flow (5m candle) Latest	Net buying pressure in the most recent candle	<code>2 × taker_buy_base_volume - volume</code> (positive = net buying)
... Selected Range Sum	Aggregated totals over the Grafana time range	<code>SUM()</code> of the corresponding metric
... Selected Range Avg	Per-candle average over the Grafana time range	<code>AVG()</code> of the corresponding metric
Buy Share % Selected Range Weighted Avg	Volume-weighted average buy share	<code>SUM(taker_buy_base_volume) / SUM(volume)</code>

Time series charts

Panel	Chart type	What it shows
Buy vs Sell Volume	Mirrored bars from zero (green up / red down)	Per-candle buy and sell volume. Sell volume is negated so both series extend from a shared zero baseline. Visually highlights which side dominates each candle.
Buy Share % vs Sell Share %	Stacked area (green + red = 100%)	Each candle's volume split as a proportion. The boundary line between green and red is the buy share ratio. A rising green area means buyers are capturing more of each candle's volume.
Net Taker Flow & Rolling Net Flow	Dual line (grey + yellow)	Grey: raw per-candle net flow. Yellow: 24-candle (2h) rolling sum. The rolling line smooths noise and reveals the trend direction of net buying pressure.
Flow vs Inventory Divergence (Multi-axis)	Multi-axis compound line	Three series on separate axes — Net Flow (grey, left), Inventory Δ (orange, right), Divergence (purple, left). Divergence = Net Flow – Inventory Δ. When divergence grows, spot buying/selling is not being reflected in the margin pool, suggesting off-exchange or spot-only activity.

PIBRBS row

Overlays five core series on shared time axes: **Price**, **Inventory**, **Borrow proxy**, **Repay proxy**, **Buy/Sell % change**.

Panel	Series	What it shows
Combined: Price, Inventory, Borrow/Repay Proxy, Buy/Sell Change	All five	Master view. Price (green line, left axis), Inventory (orange line, right axis), Borrow proxy (red bars down from zero, right axis), Repay proxy (blue bars up from zero, right axis), Buy/Sell Log Change (thin grey line, left axis).
Price & Buy/Sell Change	Price + Buy/Sell Log Change	Isolates the relationship between price movement and shifts in the buy-to-sell ratio. Buy/Sell Log Change = $\ln(\text{ratio} / \text{prev_ratio}) \times 100$ where ratio = $\text{taker_buy_base_volume} / \text{sell_volume}$. Uses natural log instead of percentage change for symmetric scaling — equal-magnitude moves in opposite directions produce equal-magnitude values (e.g. ratio doubling = +69, halving = -69). Outlier spikes remain visible but no longer distort the axis asymmetrically.
Price, Inventory & Borrow/Repay Proxy	Price + Inventory + Borrow/Repay	Isolates the supply-side picture. Borrow proxy = negative inventory drops (someone borrowed). Repay proxy = positive inventory jumps (someone repaid).

Series definitions

Series	Source	Formula
Price	spot_klines.close	Raw close price
Inventory	margin_available_inventory_snapshots.available_inventory	Raw pool supply
Borrow proxy	Inventory snapshots	$-\text{MAX}(0, \text{prev_inventory} - \text{inventory})$ — shown as red bars below zero
Repay proxy	Inventory snapshots	$\text{MAX}(0, \text{inventory} - \text{prev_inventory})$ — shown as blue bars above zero
Buy/Sell Log Change	spot_klines	$\ln(\text{ratio} / \text{prev_ratio}) \times 100$ where ratio = $\text{taker_buy_base_volume} / (\text{volume} - \text{taker_buy_base_volume})$. Uses log-change instead of percentage change: symmetric around zero, no upper-bound asymmetry. A value of +69 means the buy/sell ratio doubled; -69 means it halved. Small changes (±5) are approximately equal to percentage changes.

Directional Margin Map row

Separates **base-side borrowing** from **quote-side borrowing** to resolve the directional ambiguity in the original inventory panels. For BTCUSDC: BTC inventory drawdown suggests short pressure (base borrowed to sell), while USDC inventory drawdown suggests long pressure (quote borrowed to buy BTC). Uses a hidden `quote_asset` template variable auto-resolved from the selected symbol.

Panel	Chart type	What it shows
Price, Base Inventory & Quote Inventory	Multi-axis lines (3 Y axes: price left, base inv right, quote inv right)	Price (green), base asset inventory (orange), quote asset inventory (blue). Each axis has independent scale in raw units. Divergence between base and quote drawdowns reveals directional leverage bias.
Base Borrow / Repay Proxy	Mirrored bars from zero (red down / blue up)	Same LAG-based diff as PIBRBS but isolated to the base asset (e.g. BTC). Red bars = base borrowed (short pressure). Blue bars = base repaid (short covering).
Quote Borrow / Repay Proxy	Mirrored bars from zero (dark orange down / cyan up)	Same logic for the quote asset (e.g. USDC). Dark orange bars = quote borrowed (long pressure). Cyan bars = quote repaid (long unwind / deleveraging).
Directional Stress Ratio	Oscillator centered at zero (continuous color scale)	<code>quote_stress_z - base_stress_z</code> . Computed from raw inventory using 288-point rolling z-scores. Positive = quote borrow pressure dominates → long bias . Negative = base borrow pressure dominates → short bias .

Directional signal interpretation

Pattern	Meaning
Base inventory falling, quote stable	Base being borrowed → short-biased pressure
Quote inventory falling, base stable	Quote being borrowed to buy base → long-biased pressure
Both falling	Both sides leveraging — high-stress environment, watch for liquidation cascades
Both recovering	Deleveraging across both sides — selling/covering pressure may be exhausting
Directional Stress > +1	Strong quote-side stress → high long bias
Directional Stress < -1	Strong base-side stress → high short bias

Quote Flow & Z-scores row

Adds quote-denominated flow metrics and statistical normalization for cross-asset comparability and outlier detection.

Panel	Chart type	What it shows
Net Taker Flow & Rolling Net Flow (Quote Terms)	Dual line (grey + yellow)	Same structure as the base-unit net flow panel but in quote currency (e.g. USDC). Formula: <code>2 × taker_buy_quote_volume - quote_volume</code> . Makes cross-asset flow comparison meaningful.

Panel	Chart type	What it shows
Net Flow Z-score (24h rolling)	Oscillator line centered at zero	$(\text{net_flow} - \text{rolling_mean}) / \text{rolling_std}$ over a 288-candle (24h) window. Values $> +2$ = unusually aggressive buying. Values < -2 = unusually aggressive selling.
Volume Shock (24h Z-scores)	Three bar series	Z-score of total volume (grey), buy volume (green), sell volume (red). All on 24h rolling window. Spikes identify candles with statistically significant volume — filters noise from low-volume oscillation.
Flow Regime Shift	Multi-axis compound (4 series, 2 Y axes)	Left axis: raw net flow (grey) + 2h rolling net flow (yellow, thick). Right axis: buy share % (green) + buy/sell log change (orange). Shows when spot flow flips from seller-dominant to buyer-dominant or vice versa.

Signals for price movement prediction

Primary signal: inventory drawdown diverging from price

The strongest predictive setup occurs when the **Normalized Comparison** panel shows inventory falling while price holds steady or rises. This indicates hidden leverage buildup — traders are borrowing aggressively to go long. The larger the divergence, the more fragile the structure:

- **Inventory drawdown + price stable/rising** → Crowded leveraged long. A price dip can trigger cascading liquidations, amplifying the sell-off.
- **Inventory recovering + price falling** → Deleveraging in progress. Forced closures return supply to the pool. Selling pressure may be exhausting.

Important directional caveat: The above applies to the **combined** (base-only) inventory view. For precise direction, use the **Directional Margin Map** panels which separate base vs quote:

- **Base inventory drawdown** (e.g. BTC pool draining) = base being borrowed, usually **short-biased** pressure
- **Quote inventory drawdown** (e.g. USDC pool draining) = quote being borrowed to buy base, usually **long-biased** pressure
- A combined inventory drawdown with price rising is only reliably a "crowded long" signal when the **quote-side** drawdown dominates (check the Directional Stress Ratio)

Correlation regime reading

Correlation value	Meaning	Implication
Strongly negative (< -0.5)	Price up → inventory down (or vice versa)	Active margin borrowing against the trend. Classic stress buildup phase.
Near zero (-0.2 to +0.2)	No systematic relationship	Neutral — margin activity is not directionally aligned with price.
Strongly positive (> +0.5)	Price and inventory move together	Deleveraging or re-collateralization — positions being unwound. Often seen during or after liquidation cascades.

Watch for **correlation sign flips** — a shift from negative to positive often marks the transition from leverage buildup to forced unwind.

Stress proxy thresholds

Stress Proxy	Regime	Interpretation
< 0.5	LOW	Normal pool supply. No unusual borrowing pressure.
0.5 – 1.5	MEDIUM	Elevated borrowing. Pool is thinning. Monitor closely.
> 1.5	HIGH	Significant supply depletion. Pool is ≥1.5 standard deviations below its rolling mean. Conditions are fragile.

Config change events as leading indicators

Exchange-initiated changes to margin tiers, collateral ratios, or liquidation parameters often **precede volatility**. When Binance tightens risk parameters (raises liquidation ratios, lowers max borrowable), it typically signals the exchange sees elevated risk before it is reflected in price.

Combining signals

The highest-conviction setup combines multiple dashboard elements:

1. **Stress regime** at MEDIUM or HIGH
2. **Inventory drawdown** visible in the normalized comparison (diverging below price)
3. **Rolling correlation** deeply negative (leverage building against trend)
4. **Config change** annotation appearing (exchange tightening parameters)

This combination suggests a fragile market structure where a relatively small adverse price move can trigger outsized liquidation cascades.

Money Flow Lite signals

Net Taker Flow direction

Condition	Reading	Trade implication
Net Flow persistently positive	Taker buyers dominate candle after candle	Spot demand is genuine. Supports long bias.
Net Flow persistently negative	Taker sellers dominate	Sustained selling pressure. Supports short bias or staying flat.
Net Flow flipping from positive to negative	Buyer exhaustion	Momentum shift — consider closing longs or initiating shorts.
Net Flow flipping from negative to positive	Seller exhaustion	Potential reversal — consider closing shorts or initiating longs.

Rolling Net Flow (yellow line) as trend filter

The 24-candle (2h) rolling sum acts as a momentum smoothing filter:

- **Rolling line rising and above zero** → Sustained buying wave. Trend favors longs.
- **Rolling line falling and below zero** → Sustained selling wave. Trend favors shorts.
- **Rolling line crossing zero** → Momentum regime change. Key signal for entry/exit timing.

Buy Share % reading

Buy Share	Meaning
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Buy Share	Meaning
> 55% sustained	Buyers are aggressive. Bullish flow imbalance.
< 45% sustained	Sellers are aggressive. Bearish flow imbalance.
Oscillating 48–52%	Balanced market — no directional edge from flow alone.

A sudden spike in buy share from a balanced baseline (e.g. 50% → 60%) often precedes a short-term price move up, and vice versa.

Flow vs Inventory Divergence

This is the most actionable panel in Money Flow Lite because it connects spot-side activity (net taker flow) with margin-side activity (inventory changes).

Pattern	Interpretation	Trade implication
Divergence rising (purple up)	Net buying exceeds inventory recovery	Spot demand is outpacing margin pool replenishment. Bullish pressure building. Favors long.
Divergence falling (purple down)	Net selling exceeds inventory drawdown	Spot selling is outpacing borrowing. Bearish pressure. Favors short.
Divergence near zero	Flow and inventory changes are balanced	No edge from this signal.
Net Flow positive + Inventory Δ negative	Buyers on spot while borrowers draw down the pool	Double bullish — both spot and margin participants are positioned long. Strong long signal, but also fragile if sentiment flips.
Net Flow negative + Inventory Δ positive	Sellers on spot while borrowers repay	Double bearish unwind — deleveraging plus spot selling. Continuation of downtrend likely.

PIBRBS signals

Buy/Sell Log Change (grey line) as momentum oscillator

This series measures the candle-over-candle log-change in the buy-to-sell ratio ($\ln(\text{ratio} / \text{prev_ratio}) \times 100$). It functions as a high-frequency momentum oscillator with symmetric scaling — equal moves up and down produce equal magnitudes.

- **Spikes above +5** → Sudden shift toward buyers. Short-term bullish.
- **Spikes below -5** → Sudden shift toward sellers. Short-term bearish.
- **Large spikes (±50 to ±300+)** → Extreme outlier candles where buy/sell balance shifted dramatically. These remain fully visible and indicate candles where one side dominated volume almost entirely.
- **Sustained drift in one direction** → Trending flow imbalance. Stronger signal than isolated spikes.

Reference values: +69 ≈ ratio doubled (2× more buyers), -69 ≈ ratio halved, +230 ≈ 10× shift, -230 ≈ 0.1× shift.

Borrow/Repay proxy reading

Pattern	Meaning	Trade implication
Cluster of red (borrow) bars	Rapid inventory drawdown — margin borrowing accelerating	Leveraged positions building. If price is rising, crowded long. If price is falling, short-sellers borrowing to add.

Pattern	Meaning	Trade implication
Cluster of blue (repay) bars	Inventory recovering — positions being closed or collateral returned	Deleveraging phase. Selling pressure from forced closures may be near exhaustion.
Borrow bars + price rising	Traders borrowing to go long (or short-sellers being squeezed into repaying)	Fragile bullish — works until it doesn't. Watch for repay bars signaling the unwind.
Borrow bars + price falling	Traders borrowing to short (or longs being liquidated)	Bearish momentum reinforced by leverage.
Repay bars + price falling	Forced unwind of longs	Liquidation cascade may be underway. Look for repay bars to slow as a bottom signal.
Repay bars + price rising	Short-sellers closing (covering)	Short squeeze dynamics. Bullish until repay activity subsides.

Cross-dashboard signal synthesis

The strongest setups combine signals from both the **Inventory** (main) dashboard and the **Flow** (PIBRBS) dashboard:

High-conviction long setup

1. **Inventory dashboard:** Stress regime LOW or MEDIUM, inventory stable or recovering
2. **Flow dashboard:** Net Taker Flow persistently positive, rolling net flow above zero and rising
3. **Flow dashboard:** Buy Share % > 55%
4. **PIBRBS:** Buy/Sell Log Change trending positive, no cluster of borrow bars (leverage not excessive)
5. **Flow dashboard:** Divergence (purple) rising — spot demand exceeding inventory change

This combination shows genuine spot-driven demand without excessive leverage, the healthiest form of bullish pressure.

High-conviction short setup

1. **Inventory dashboard:** Stress regime MEDIUM or HIGH, inventory drawdown accelerating
2. **Flow dashboard:** Net Taker Flow persistently negative, rolling net flow below zero and falling
3. **Flow dashboard:** Buy Share % < 45%
4. **PIBRBS:** Cluster of borrow bars (red) while price rises — leverage building on the long side
5. **Inventory dashboard:** Rolling correlation deeply negative (leverage diverging from price)

This combination shows leveraged longs building against weakening spot demand — the classic setup for a liquidation cascade.

Reversal detection

Watch for these transitions across dashboards:

- **Long → short reversal:** Rolling net flow crosses below zero + repay bars appear (longs unwinding) + stress proxy rising → bearish shift
- **Short → long reversal:** Rolling net flow crosses above zero + borrow bars stop (selling pressure exhausted) + divergence flipping positive → bullish shift
- **Confirmation:** On the Inventory dashboard, a correlation sign flip (negative → positive) often confirms the regime change

Directional margin map signals

Use the **Directional Margin Map** row on the Flow dashboard for precise long/short bias:

- **Directional Stress Ratio rising above +1:** Quote pool under stress → long-biased leverage building. Confirm with rolling net flow positive and buy share > 52%.
- **Directional Stress Ratio falling below -1:** Base pool under stress → short-biased leverage building. Confirm with rolling net flow negative and buy share < 48%.
- **Quote borrow cluster + base repay cluster:** Short-to-long rotation — shorts covering (base repaid) while new longs open (quote borrowed).
- **Base borrow cluster + quote repay cluster:** Long-to-short rotation — longs unwinding (quote repaid) while new shorts open (base borrowed).
- **Net Flow Z-score > +2 with quote borrow bars:** Strong conviction long signal — statistically unusual buying pressure backed by leveraged demand.
- **Net Flow Z-score < -2 with base borrow bars:** Strong conviction short signal — statistically unusual selling pressure backed by leveraged shorting.
- **Volume Shock spike (any z-score > 3):** Unusually large candle — check whether buy or sell z-score dominates to determine direction of the shock.

On-chain data sources for next price move detection

Margin inventory analysis (this system) tracks **CEX lending pool** activity. On-chain analytics adds a second layer: **actual capital movements recorded on the blockchain** — wallet flows, exchange deposits/withdrawals, holder behavior, miner activity. These are complementary layers with different lead times and signal types.

Signal layers and their lead times

Layer	Examples	Typical lead time	What it measures
On-chain holder flows	Exchange inflow/outflow, SOPR, MVRV	Hours to days	Accumulated capital positioning of all market participants
Derivatives structure	OI, funding rates, liquidation maps	Minutes to hours	Leverage positioning and crowdedness
Margin pool inventory	This system	Minutes	CEX lending demand (this system's core edge)
Spot taker flow	Net flow, buy share %	Real-time	Immediate execution-side pressure

The highest-conviction setups stack signals from all four layers.

Tier 0: All-in-one platform (consider first)

Alphractal (<https://alphractal.com> · API: <https://api.alphractal.com>)

Best for: Replacing Glassnode + CryptoQuant + Coinglass with a single API. Covers on-chain, derivatives, sentiment, macro, and market data under one key. Free to start.

From the live API spec, confirmed categories:

Category	Count	Notable metrics
On-chain	300+	SOPR, LTH/STH supply, address balance distributions, exchange flows, UTXO age bands
Derivatives	60+	OI, funding rates, liquidations, long/short ratios, perpetual futures
Mining	+ proprietary	Hash rate, difficulty, Puell Multiple + <code>capitulation_oscillator</code> , <code>hash_momentum_score</code> , <code>mining_equilibrium</code> , <code>network_stress</code>

Category	Count	Notable metrics
Sentiment	80+	Fear & Greed, Google Trends, Twitter/X, 4chan
Macro	300+	DXY, S&P 500, Treasury yields, global liquidity — unique, no other platform bundles this
Proprietary models	23	Pi Cycle, CVDD, Reserve Risk, Accumulation Heatmaps, smart_money_flow
Screener	100+ chart types	Per-asset OI trends, buy/sell volume by timeframe, LSR, MA distances

API: REST, OpenAPI 3.1. Auth via **X-API-Key** header. Pattern: **GET**
<https://api.alphractal.com/{asset}/{category}/{metric}?startDate=...&endDate=...>

Advantages over the Tier 1 stack below:

- One API replaces Glassnode + CryptoQuant + Coinglass (users explicitly report switching from all three)
- Macro data (DXY, rates) bundled — none of the specialist platforms offer this
- Proprietary derived signals already computed — **capitulation_oscillator** etc. would require raw data + computation from Glassnode
- 1,000+ assets — broader than Glassnode/CryptoQuant which focus on BTC/ETH
- AI layer (Alpha AI) for natural-language querying of all 1,500+ metrics
- Free tier — lower barrier to prototype than Glassnode's paid-only historical data

Weaknesses:

- Data provenance less transparent than Glassnode (which publishes full methodology per metric)
- Newer platform — data quality edge cases harder to verify vs Glassnode's 7-year track record
- Liquidation heatmap visual tool not as mature as Coinglass's UI
- No entity-level wallet intelligence (Arkham's territory)

Verdict: Best single starting point for the on-chain data extension. Prototype with the free tier. Supplement with Glassnode only if you need auditable methodology for specific metrics (SOPR, LTH/STH) in production.

Tier 1 APIs — systematic quantitative analytics

Glassnode (<https://glassnode.com>)

Best for: Comprehensive BTC and ETH on-chain analysis. The most data-complete platform for systematic signal construction.

Metric	API path example	Signal logic	Lead time
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Metric	API path example	Signal logic	Lead time
Exchange Net Position Change	/v1/metrics/distribution/exchange_net_position_change	Net BTC moving to/from all exchanges. Sustained outflow (negative) = accumulation → bullish. Inflow spike = distribution → bearish.	6–48h
SOPR (Spent Output Profit Ratio)	/v1/metrics/indicators/sopr	>1 = average moved coin is in profit (sellers may realize gains). <1 = selling at a loss (capitulation — often near bottoms).	12–72h
Short-Term Holder SOPR	/v1/metrics/indicators/sopr_less_155d	STH SOPR dipping below 1 then recovering = local bottom signal. Staying above 1 after a rally = healthy trend.	2–24h
Long-Term Holder SOPR	/v1/metrics/indicators/sopr_more_155d	LTH SOPR below 1 = long-term holders capitulating (rare, strong bottom signal). >1 with high values = distribution phase.	Days to weeks

Metric	API path example	Signal logic	Lead time
MVRV Z-Score	/v1/metrics/market/mvrv_z_score	Market cap vs Realized cap normalized by std. >7 = historically overvalued (past tops). <0 = historically undervalued (past bottoms).	Weeks (structural)
Exchange Stablecoin Ratio	/v1/metrics/distribution/exchange_stablecoin_supply_ratio	High stablecoin reserves on exchanges relative to BTC market cap = dry powder for buying. Rising = bullish dry powder building.	1-7 days
Estimated Leverage Ratio	/v1/metrics/derivatives/futures_estimated_leverage_ratio	OI / exchange reserves. High = system is over-leveraged → fragile, prone to cascades.	Hours

API: REST (https://api.glassnode.com/v1/metrics/{category}/{metric}?a=BTC&api_key=...). Requires API key. Tier 1 plan (~\$39/mo) covers most of the above.

Coinglass (<https://coinglass.com>)

Best for: Derivatives micro-structure — liquidation maps, OI aggregation, funding rates. Free tier is highly functional.

Metric	What it shows	Signal logic
Liquidation Heatmap	Price levels with concentrated liquidation clusters	Price approaching a cluster = likely acceleration through that level (engine for cascade or squeeze). Critical for entry/exit placement.

Metric	What it shows	Signal logic
Open Interest	Aggregate OI across all major exchanges	OI rising with price = healthy. OI rising with flat price = leverage buildup → fragile. OI dropping with price = forced deleveraging (capitulation or correction).
Funding Rate	Per-exchange and aggregated 8h funding	Persistent positive (>0.05% per 8h) = longs paying heavily → crowded. Will mean-revert via flush. Persistent negative = shorts crowded → squeeze setup.
Long/Short Ratio	% of accounts long vs short on Binance futures	Extreme long ratio (>70%) = contrarian bearish. Extreme short ratio (<30% long) = contrarian bullish.

API: <https://open-api.coinglass.com/public/v2/...> — free tier with rate limits. Premium unlocks higher resolution.

CryptoQuant (<https://cryptoquant.com>)

Best for: Exchange-specific inflow/outflow, miner behavior, stablecoin supply.

Metric	Signal logic
Exchange Inflow (all exchanges)	Large sudden inflows (BTC moving onto exchanges) = intent to sell. Often precedes selling within 1–6h.
Exchange Outflow	Sustained outflow = holders withdrawing to self-custody (accumulation). Bullish medium-term.
Miner Reserve	Miners reducing reserves (selling) = supply pressure. Often front-runs volatility.
Stablecoin Exchange Inflow	Stablecoins moving onto exchanges = fresh buying power arriving. Short-term bullish.

API: REST, requires subscription.

Tier 2 APIs — specialized or freemium

Arkham Intelligence (<https://arkhamintelligence.com>)

Best for: Entity-level intelligence. Maps blockchain addresses to real-world entities (exchanges, funds, market makers, known whales). Best used as a surveillance complement, not a systematic signal source.

Use case	What it enables
Whale tracking	Alert when labeled wallets (Jump, known miners, OTC desks) move large amounts
Exchange cold wallet monitoring	Track exchange reserve movements at the wallet level, not just aggregate
Smart money alerts	When known profitable addresses accumulate or distribute

Comparison vs Glassnode: Glassnode excels at systematic, time-series, statistically normalizable signals. Arkham excels at surveillance and actor-specific intelligence. They answer different questions — Glassnode answers "what is the aggregate doing?", Arkham answers "what is *this specific actor* doing?" For algorithmic trading setups, Glassnode is more directly actionable; Arkham adds context.

Whale Alert (<https://whale-alert.io>)

Real-time alerts for transactions >\$X on-chain. Simple noise layer — useful as a trigger for checking deeper data, not as a signal itself. API + Telegram bot.

Nansen (<https://nansen.ai>)

Best for Ethereum, ERC-20, and DeFi analytics. Smart money wallet labels (funds, DEX whales). Less relevant for BTC-centric analysis. Strong for altcoin smart-money positioning.

Priority metrics to fetch — ranked by signal quality

Short-term (1–8 hours)

1. **Exchange large inflow alert** (Glassnode / CryptoQuant) — spike of >500 BTC arriving on exchange in 1h often precedes selling within 1–6h
2. **Funding rate** (Coinglass) — rate >+0.10% per 8h signals crowded longs → flush imminent
3. **Liquidation heatmap proximity** (Coinglass) — price within 1% of a major liquidation cluster means the cluster is the next magnetic target
4. **Stablecoin exchange inflow** (CryptoQuant) — fresh USDC/USDT arriving = buying fuel → short-term bullish
5. **Volume Shock Z-score** (already collected) — statistically unusual candle validates other on-chain signals

Mid-term (1–7 days)

1. **Exchange Net Position 7d trend** (Glassnode) — sustained outflow = structural accumulation → medium-term bullish
2. **SOPR 7d trend** (Glassnode) — sustained >1 = distribution. Sustained <1 = capitulation exhausting
3. **LTH SOPR** (Glassnode) — LTH capitulating (<1) = strong bottom signal; LTH distributing heavily = top signal
4. **OI trend relative to price** (Coinglass) — divergence between OI growth and price growth = leverage buildup
5. **Miner reserve 7d change** (CryptoQuant) — large miner outflow → supply increase incoming

How to integrate into this system

Database schema extension

Add an `onchain_metrics` table using the same pattern as `derived_metrics`:

```
CREATE TABLE onchain_metrics (
  id          BIGSERIAL,
  collected_at TIMESTAMPTZ NOT NULL,
  asset       TEXT          NOT NULL,
  metric_name TEXT          NOT NULL,
  metric_value DOUBLE PRECISION,
  source      TEXT          NOT NULL, -- 'glassnode', 'coinglass', etc.
  window_label TEXT,
  metadata    JSONB
);
```

Correlation approach

- Z-score normalize all on-chain signals over a 24h/7d window (same as net flow Z-scores already in this system)
- Plot each signal alongside spot price on a shared time axis

- Compute **lead correlation**: correlate signal at time t against price return at $t+N$ for $N = 1h, 4h, 12h, 24h, 48h$ to find optimal leading window
- Prioritize signals with statistically significant lead correlation at $N > 0$ (predictive rather than coincident)

Recommended starting stack

Data source	What to add	Why
Alphractal (start here)	Exchange flows + SOPR + OI + Funding Rate + Macro (DXY)	Single API covering Glassnode + CryptoQuant + Coinglass. Free tier to prototype.
Glassnode (optional upgrade)	Same metrics as Alphractal but with published methodology	Add if you need auditable data provenance for SOPR/LTH signals in a production system
Coinglass (optional)	Liquidation heatmap visual UI	Add only if you need the visual heatmap tool; the underlying data is in Alphractal's API
This system	Margin inventory + spot flow (already running)	CEX lending demand — the existing edge not available from any on-chain API

Arkham is complementary for entity/whale surveillance (different problem from aggregate analytics). Nansen is worth adding if you trade ETH/DeFi tokens.